Proving Angle Relationships — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see "Your turn."

Name: _____ Date: _____

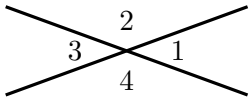
☞ *Video: Four facts you can use*

Key Idea

Linear Pair Postulate	a linear pair is **supplementary** (adds to $180°$)
Vertical Angles Congruence Theorem	vertical angles are **congruent**
Right Angles Congruence Theorem	all right angles are **congruent**
Congruent Supplements / Complements	two angles supplementary to the **same** angle are congruent

☞ *Video: Why vertical angles are congruent*

Watch Me — Worked Example



$m\angle 1 + m\angle 2 = 180°$ **Linear Pair Postulate**

$m\angle 3 + m\angle 2 = 180°$ **Linear Pair Postulate**

$\angle 1 \cong \angle 3$ **Congruent Supplements Theorem**

☞ *Video: Use it to find angles*

Watch Me — Worked Example

Same picture: $m\angle 1 = 3x + 10,\ m\angle 3 = 5x - 30.$

$3x + 10 = 5x - 30$ vertical angles are **congruent**

$40 = 2x,\ x =$ 20 solve

$m\angle 1 = 3(20) + 10 =$ $70°$

$m\angle 2 = 180 - 70 =$ $110°$ $\angle 1$ and $\angle 2$ are a **linear pair**

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☞ *Video: Watch out*

Common Mistake

Vertical angles are **across** from each other — they are equal. Angles **next to** each other on a line are a linear pair — they add to $180°$, they are **not** equal.

☞ *Video: Your turn*

Your Turn

$\angle A$ and $\angle B$ are both supplementary to $\angle C$. $m\angle A = 2x + 15,\ m\angle B = 3x - 5$. Find x and $m\angle A$.

Congruent **Supplements**: $2x + 15 = 3x - 5 \Rightarrow x =$ 20, $m\angle A =$ $55°$

Notice

Remember: Linear pair: adds to $180°$. Vertical angles: congruent. All right angles are congruent. Supplementary (or complementary) to the **same** angle $\Rightarrow$ congruent.

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